

Technology Stack

Date	11 July 2006
Team ID	
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3, Bootstrap	Used to create a simple, responsive, and user-friendly web interface for entering HDI indicator values and displaying prediction results.

2	Backend / Server-Side <i>(e.g., Node.js, Python/Django, Java)</i>	<i>Python, Flask</i>	<i>Flask is a lightweight Python web framework used to process user requests, integrate the Machine Learning model, and return prediction results efficiently</i>
3	Database / Data Storage <i>(e.g., MySQL, MongoDB, PostgreSQL)</i>	<i>CSV Dataset (UNDP/World Bank/Kaggle)</i>	<i>The project uses a CSV file to store HDI data for model training and testing. It is easy to manage and suitable for academic Machine Learning projects.</i>

4	Cloud / Hosting / Deployment <i>(e.g., AWS, Azure, Heroku, Docker)</i>	<i>Render (or Local Flask Server)</i>	<i>Render provides simple cloud deployment, while Flask's local server is used during development and testing</i>
5	Version Control & CI/CD	<i>Git & GitHub</i>	<i>Git is used for version control, and GitHub stores the project repository, enabling collaboration, backup, and future updates.</i>
6	Third-Party APIs / Other Tools <i>(e.g., Stripe, Twilio, Google Maps)</i>	<i>Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Pickle</i>	Scikit-learn is used for Machine Learning, Pandas and NumPy for data processing, Matplotlib and Seaborn for visualization, and Pickle for saving/loading the trained model.